Lecture 7: Sections 2.1 and 2.2

Rate of change

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Tangent Lines: Definition

Definition

The **tangent line** to a curve $y = f(x)$ at a point $P(a, f(a))$ is the line that:

- Passes through the point $P(a, f(a))$
- Has slope equal to the **instantaneous rate of change** of f at $x = a$
- Best approximates the curve near point P

Slope of Tangent Line

The slope of the tangent line at $x = a$ is given by:

$$m_{\tan} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

This is also called the **derivative** of f at $x = a$, denoted $f'(a)$.

Equation of Tangent Line

Point-Slope Form

Once we have the slope $m_{\tan} = f'(a)$, the equation of the tangent line at $P(a, f(a))$ is:

$$y - f(a) = f'(a)(x - a)$$

Step-by-Step Procedure

1. Find the point of tangency: $(a, f(a))$
2. Compute the slope: $m = f'(a)$
3. Use point-slope form: $y - f(a) = m(x - a)$
4. Simplify to slope-intercept form if needed

Worked Example: Finding Tangent Line Equation

Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

1. Point of tangency: $f(3) = 3^2 = 9$, so point is $(3, 9)$
2. Compute slope using limit definition:

$$\begin{aligned} f'(3) &= \lim_{h \to 0} \frac{f(3+h) - f(3)}{h} \\ &= \lim_{h \to 0} \frac{(3+h)^2 - 9}{h} \\ &= \lim_{h \to 0} \frac{9 + 6h + h^2 - 9}{h} \\ &= \lim_{h \to 0} \frac{6h + h^2}{h} \\ &= \lim_{h \to 0} (6 + h) = 6 \end{aligned}$$

3. Equation: $y - 9 = 6(x - 3)$
4. Simplified: $y = 6x - 9$

Worked Example: Computing Slope

Find the slope of the tangent line to $f(x) = \sqrt{x}$ at $x = 4$.

$$
\begin{aligned}
f'(4) &= \lim_{h \to 0} \frac{f(4+h) - f(4)}{h} \\
&= \lim_{h \to 0} \frac{\sqrt{4+h} - 2}{h} \\
&= \lim_{h \to 0} \frac{\sqrt{4+h} - 2}{h} \cdot \frac{\sqrt{4+h} + 2}{\sqrt{4+h} + 2} \\
&= \lim_{h \to 0} \frac{(4+h) - 4}{h(\sqrt{4+h} + 2)} \\
&= \lim_{h \to 0} \frac{h}{h(\sqrt{4+h} + 2)} \\
&= \lim_{h \to 0} \frac{1}{\sqrt{4+h} + 2} = \frac{1}{4}
\end{aligned}
$$

Rates of Change and Velocity

The average rate of change of f over $[a, a+h]$ is:

$$\frac{f(a+h)-f(a)}{h}$$

The instantaneous rate of change at $x = a$ is:

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = f'(a)$$

If $s(t)$ represents position at time t, then:

- Average velocity over $[t, t+h]$: $\dfrac{s(t+h)-s(t)}{h}$
- Instantaneous velocity at time t: $s'(t)$

Worked Example: Computing Velocity

The position of an object is given by $s(t) = -16t^2 + 100t + 5$ (in feet). Find the instantaneous velocity at $t = 2$ seconds.

$$
\begin{aligned}
s'(2) &= \lim_{h \to 0} \frac{s(2+h) - s(2)}{h} \\
s(2+h) &= -16(2+h)^2 + 100(2+h) + 5 \\
&= -16(4 + 4h + h^2) + 200 + 100h + 5 \\
&= -64 - 64h - 16h^2 + 200 + 100h + 5 \\
&= 141 + 36h - 16h^2 \\
s'(2) &= \lim_{h \to 0} \frac{(141 + 36h - 16h^2) - 141}{h} \\
&= \lim_{h \to 0} \frac{36h - 16h^2}{h} \\
&= \lim_{h \to 0} (36 - 16h) = 36 \text{ ft/s}
\end{aligned}
$$

Practice Problems: Tangent Lines

Basic Problems

1. Find the equation of the tangent line to $f(x) = 3x^2 - 2x + 1$ at $x = 1$
2. Find the slope of the tangent line to $f(x) = \frac{1}{x}$ at $x = 2$
3. Find the equation of the tangent line to $f(x) = x^3$ at $x = -1$

Applied Problems

1. The position function $s(t) = t^2 + 4t$ gives position in meters. Find the instantaneous velocity at $t = 3$ seconds.
2. The revenue function is $R(x) = 50x - 0.1x^2$. Find the instantaneous rate of change of revenue when producing 100 units.

Definition of Derivative Function

The **derivative** of a function f is the function f' defined by:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

for all x where this limit exists.

- Leibniz notation: $\frac{dy}{dx}$ or $\frac{d}{dx}[f(x)]$
- Prime notation: $f'(x)$
- The derivative represents the **instantaneous rate of change** of f at any point x
- $f'(a)$ = slope of tangent line at $x = a$
- $f'(a)$ = instantaneous rate of change at $x = a$
- $f'(a)$ = velocity if f represents position

Finding Derivative Using Definition

Find the derivative of $f(x) = x^2 + 3x$ using the definition.

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{[(x+h)^2 + 3(x+h)] - [x^2 + 3x]}{h} \\
&= \lim_{h \to 0} \frac{[x^2 + 2xh + h^2 + 3x + 3h] - [x^2 + 3x]}{h} \\
&= \lim_{h \to 0} \frac{2xh + h^2 + 3h}{h} \\
&= \lim_{h \to 0} \frac{h(2x + h + 3)}{h} \\
&= \lim_{h \to 0} (2x + h + 3) = 2x + 3
\end{aligned}
$$

Worked Example: Rational Function

Find the derivative of $f(x) = \dfrac{1}{x}$ using the definition.

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\
&= \lim_{h \to 0} \frac{\frac{x-(x+h)}{x(x+h)}}{h} \\
&= \lim_{h \to 0} \frac{\frac{-h}{x(x+h)}}{h} \\
&= \lim_{h \to 0} \frac{-1}{x(x+h)} \\
&= -\frac{1}{x^2}
\end{aligned}
$$

Computing Instantaneous Velocity

The height of a projectile is given by $h(t) = -16t^2 + 64t + 80$ feet. Find the velocity function and compute the velocity at $t = 2$ seconds.

$$
\begin{aligned}
v(t) = h'(t) &= \lim_{h \to 0} \frac{h(t+h) - h(t)}{h} \\
&= \lim_{h \to 0} \frac{[-16(t+h)^2 + 64(t+h) + 80] - [-16t^2 + 64t + 80]}{h} \\
&= \lim_{h \to 0} \frac{[-16(t^2 + 2th + h^2) + 64t + 64h + 80] + 16t^2 - 64t - 80}{h} \\
&= \lim_{h \to 0} \frac{-32th - 16h^2 + 64h}{h} \\
&= \lim_{h \to 0} (-32t - 16h + 64) = -32t + 64
\end{aligned}
$$

At $t = 2$: $v(2) = -32(2) + 64 = 0$ ft/s (at maximum height)

Practice Problems: Definition of Derivative

Basic Derivative Problems

1. Use the definition to find the derivative of $f(x) = 4x - 7$
2. Use the definition to find the derivative of $f(x) = x^2 - 5x$
3. Use the definition to find the derivative of $f(x) = \sqrt{x}$

Applied Problems

1. The position function $s(t) = t^3 - 6t^2$ gives position in meters. Find the velocity function and determine when the object is at rest.
2. The cost function is $C(x) = 100 + 5x + 0.1x^2$. Find the marginal cost function.

Advanced Practice Problems

Challenging Derivative Problems

1. Use the definition to find the derivative of $f(x) = \dfrac{1}{x^2}$
2. Use the definition to find the derivative of $f(x) = \sqrt{x+1}$
3. The volume of a sphere is $V(r) = \frac{4}{3}\pi r^3$. Use the definition to find the rate of change of volume with respect to radius.

Theoretical Problems

1. Prove that if $f(x) = c$ (constant function), then $f'(x) = 0$ using the definition.
2. Prove that if $f(x) = mx + b$, then $f'(x) = m$ using the definition.

Key Formulas and Summary

Essential Formulas

- Derivative definition: $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$
- Tangent line equation: $y - f(a) = f'(a)(x - a)$
- Instantaneous velocity: $v(t) = s'(t)$
- Instantaneous rate of change: $f'(a)$

Problem-Solving Strategies

- For tangent lines: Find point and slope, then use point-slope form
- For derivatives: Substitute into definition and simplify
- For velocity: Differentiate position function
- Algebraic manipulation often needed: factoring, rationalizing, simplifying

Common Applications

- Physics: Velocity, acceleration
- Economics: Marginal cost, marginal revenue
- Geometry: Slope of curves
- Engineering: Rates of change in systems